

JEE MAIN REVISION MASTER: CENTER OF MASS AND COLLISION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Center of Mass Calculation & Cut-out Sections	4 PYQs	Center of Mass
Moderate	Conservation of Linear Momentum & Explosion	4 PYQs	Momentum

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Center of Mass Calculation & Cut-out Sections	$\vec{r}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i}, \quad \vec{r}_{\text{remaining}} = \frac{M\vec{r}_0 - m\vec{r}_{\text{removed}}}{M - m}$
Conservation of Linear Momentum & Explosion	$\vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}}, \quad \text{If } \vec{F}_{\text{ext}} = 0 \implies \vec{v}_{\text{cm}} = \text{const}, \quad \sum m_i \vec{v}_i = \text{const}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Center of Mass Calculation & Cut-out Sections (Moderate Yield — 4 PYQs)

Core Principle: Discrete and continuous center of mass coordinates, and negative mass method for cavity problems.

■■ **Caution / JEE Pitfall:** Hemisphere CM is at $3R/8$ from center, whereas hemispherical shell CM is at $R/2$.

■ **Speed Trick:** Treat a removed cavity as negative mass at the cavity's geometric center: $X_{\text{(new)}} = (A_1 x_1 - A_2 x_2) / (A_1 - A_2)$.

Representative Source PYQs:

- **[2025 Shift] Q#2 (MCQ):** "Three particles of mass 1kg, 2kg and 3 kg are placed at the vertices A, B and C respectively of an equilateral triangle ABC of side 1 m. T..." **[Ans: N/A]**
- **[2022 Shift] Q#3 (MCQ):** "The centre of mass of an extended body on the surface of the earth and its centre of gravity..." **[Ans: C]**
- **[2019 Shift] Q#3 (MCQ):** "Two particles which are initially at rest move towards each other under the action of their mutual attraction. If their speeds are v and \$..." **[Ans: B]**

3.2 Conservation of Linear Momentum & Explosion (Moderate Yield — 4 PYQs)

Core Principle: Motion of center of mass under zero net external force and explosive fragmentation trajectories.

■ **Caution / JEE Pitfall:** Internal forces (like explosion or spring expansion) cannot change the trajectory of the center of mass.

■ **Speed Trick:** If a projectile explodes mid-air, the center of mass continues along the original parabolic trajectory until fragments hit the ground.

Representative Source PYQs:

- **[2025 Shift] Q#1 (MCQ):** "During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the..." **[Ans: N/A]**
- **[2021 Shift] Q#1 (MCQ):** "A ball hits the floor and rebounds after an inelastic collision. In this case..." **[Ans: N/A]**
- **[2021 Shift] Q#4 (MCQ):** "A 1kg ball moving at 12ms^{-1} collides with a 2kg ball moving in opposite direction at 24ms^{-1} . If the coefficient of restitution..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the final kinetic energy of the system. II. The linear momentum is conserved. III. The kinetic energy during Δt (the collision time) is not conserved.

- (A) II and III only
- (B) I and III only
- (C) I, II and III
- (D) I and II only

Correct Answer: None

Solution Analysis: I. Final total kinetic energy equals initial II. Linear momentum is always conserved III. During collision kinetic energy gets partly converted to potential energy $\therefore$ All statements are correct

Benchmark Problem 2: KCET 2025 (Q#2)

Three particles of mass 1kg, 2kg and 3 kg are placed at the vertices A, B and C respectively of an equilateral triangle ABC of side 1 m. The centre of mass of the system from vertex A (located at origin) is

- (A) $((7/12), \frac{\sqrt{3}}{12})$
- (B) $((9/12), \frac{\sqrt{3}}{12})$
- (C) $((7/12), \frac{\sqrt{3}}{12})$
- (D) (0,0)

Correct Answer: None

Solution Analysis: $x_{\text{cm}} = (\sum m_i x_i / \sum m_i) = (1 \times 0 + 2 \times 1 + 3 \times (1/2)) / 6 = (7/12)$ $y_{\text{cm}} = (\sum m_i y_i / \sum m_i) = \frac{1 \times 0 + 2 \times 0 + 3 \times \frac{\sqrt{3}}{2}}{6} = \frac{\sqrt{3}}{4}$